Financial Time Series Analysis

Lecture 5

- TA Session Friday April 9, 2021, 11:00am - 12:00pm (EDT)
- Homework assignment due dates are being moved to Thursday

Outline

- Homework #2, problem 3
- Formal notation for AR, MA, ARMA, ARIMA
- Properties of ARMA(p,q)
- Dickey Fuller test

Invertibility of MA Processes: 1

- We have determined conditions on the coefficients of an AR(p) process for weak stationarity, namely the roots of the autoregressive polynomial, $1 - \sum_{i=1}^{p} \phi_i x^i = 0$ have modulus greater than 1.
- There is a similar condition for the coefficients of the MA(q) process. We learned that all MA(q) processes are weakly stationary.
- Consider two MA(1) processes:

$$Y_t^{(1)} = \epsilon_t - \theta_1 \epsilon_{t-1},$$

$$Y_t^{(2)} = \epsilon_t - \frac{1}{\theta_1} \epsilon_{t-1}$$

- $E(Y_t^{(i)}) = 0$ for $i = 1, 2$. While the variances are different,

Invertibility of MA Processes: 2

both processes have identical acfs, $i = 1, 2$:

$$\rho_i(h) = \begin{cases} 1 & \text{if } h = 0, \\ -\theta_1/(1+\theta_1^2) & \text{if } |h| = 1, \\ 0 & \text{if } |h| > 1. \end{cases}$$

From the point of view of the mean, variance and autocovariance, these two processes are identical. Which parametrization is preferable? We can express the residual, ϵ_t, by iterating backward:

$$\begin{aligned} \epsilon_t &= Y_t^{(1)} + \theta_1 \epsilon_{t-1} \\ &= Y_t^{(1)} + \theta_1 \big(Y_{t-1}^{(1)} + \theta_1 Y_{t-2}^{(1)} \big) \\ &= \ldots \\ &= Y_t^{(1)} + \sum_{i=1}^{\infty} \theta_1^i Y_{t-i}^{(1)}. \end{aligned}$$

Invertibility of MA Processes: 3

Similarly

$$\epsilon_t = Y_t^{(2)} + \sum_{i=1}^{\infty} \frac{1}{\theta_1^i} Y_{t-i}^{(2)}$$

.

- Now if $|\theta_1| < 1$ then the coefficients involving $Y_t^{(1)}$ are a convergent series; however, the coefficients involving $Y_t^{(2)}$ are a divergent series.
- The MA(q) process is said to be invertible
  if the residuals can be represented by an AR process with convergent coefficients.
- This is equivalent to the roots of $\theta(B) = 0$ having all roots with modulus greater than 1, i.e.

$$1 - \theta_1 z - \theta_2 z^2 - \cdots - \theta_q z^q = 0.$$

Formal Notation for AR Processes

There is a formal representation for AR(p) processes, $\{X_t\}$. Defining $Y_t = X_t - \mu$ the process satisfies

$$Y_t = \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + \epsilon_t.$$

This can be represented as

$$\phi(B) Y_t = \epsilon_t,$$

where

$$\phi(B) = (1 - \phi_1 B - \cdots - \phi_p B^p).$$

```
arparams = np.array([.30, .20, .10]) #Ar coefficients
ar = np.r_[1, -arparams] # add zero-lag and negate
ar_process = ArmaProcess(ar)
```

Formal Notation for MA Processes

Let X_t be a MA(q) with mean μ and let $Y_t = X_t - \mu$, to eliminate the mean. This can be written as

$$Y_t = \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}.$$

This can be concisely symbolized as

$$Y_t = \theta(B)\epsilon_t,$$

where

$$\theta(B) = (1 + \theta_1 B + \cdots \theta_q B^q).$$

maparams = np.array([.50,-.50]) # MA coefficients
ma = np.r_[1, maparams] # add zero-lag

Formal Notation for ARMA(p,q)

For an ARMA(p,q) model given by

$$Y_t - \phi_1 Y_{t-1} - \cdots - \phi_p Y_{t-p} = \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q},$$

we can express this using the backshift notation as

$$\phi(B) Y_t = \theta(B) \epsilon_t.$$

```
arparams = np.array([.60, -.40]) # AR coefficients
ar = np.r_[1, -arparams] # add zero-lag and negate
maparams = np.array([.50, -.50]) # MA coefficients
ma = np.r_[1, maparams] # add zero-lag
arma_process = ArmaProcess(ar, ma)
```

Formal Notation for ARIMA(p,d,q)

A process X_t is an ARMA(p,q) process if

- X_t is weakly stationary,
- $\phi(B)X_t = \theta(B)\epsilon_t, \ \forall t$ where ϵ_t is WN$(0, \sigma^2)$
- A process X_t is an ARMA(p,q) process with mean μ if Y_t is an ARMA(p,q) process where $Y_t = X_t - \mu$.
- Suppose W_t is an ARMA(p,q) process, so $\phi(B)W_t = \theta(B)\epsilon_t, \ \forall t$ where ϵ_t is WN$(0, \sigma^2)$.
- Now suppose $W_t = (1 - B)^d Y_t$, then Y_t is an ARIMA(p,d,q) process. This can be written in the notation

$$\phi(B)(1 - B)^d Y_t = \theta(B)\epsilon_t.$$

ARMA Models

ARMA Models: 1

- Last lecture we introduced and described the properties of two standard time series models::

$$AR(p) : X_t = \phi_0 + \phi_1 X_{t-1} + \cdots \phi_p X_{t-p} + \epsilon_t.$$

$$MA(q) : X_t = \mu + \theta_1 \epsilon_{t-1} + \cdots \theta_q \epsilon_{t-q} + \epsilon_t.$$

- One problem is that fitting data, especially financial data, with either a pure AR(p) model or pure MA(q) model, may require a relatively large number of parameters..
- One potential solution is to combine both into a single model, ARMA(p,q) defined by

$$X_t = \phi_0 + \phi_0 X_{t-1} + \cdots \phi_p X_{t-p} + \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}.$$

ARMA Models: 2

- For stationarity, the mean and variance must be constant. Taking expectations on both sides of the equation, we find

$$E(X_t) = \mu = \phi_0 + (\phi_1 + \cdots + \phi_p)\mu$$

or

$$E(X_t) = \mu = \frac{\phi_0}{1 - (\phi_1 + \cdots + \phi_p)}.$$

- To simplify, we set $\mu = 0$, which can be achieved by simply substituting $Y_t = X_t - \mu$.
- While this model has p+q parameters as opposed to just p or q, it is often the case that ARMA(1,1), a two-parameter model, can fit the data quite well.

ARMA(1,1): 1

The ARMA(1,1) model with $E(X_t) = \mu$ is given by

$$X_t = \phi_0 + \phi_1 X_{t-1} + \theta_1 \epsilon_{t-1} + \epsilon_t,$$

with ϵ_t is a WN$(0, \sigma_\epsilon^2)$ process.

- Assuming stationarity and taking expectations of both sides, we find $E(X_t) = \mu = \phi_0 + \phi_1 E(X_{t-1})$ or $\mu = \phi_0/(1 - \phi_1)$
- To compute the variance and autocovariance, we first note

$$\begin{aligned} Cov(X_t, \epsilon_t) &= Cov(\phi_0 + \phi_1 X_{t-1} + \theta_1 \epsilon_{t-1} + \epsilon_t, \epsilon_t) \\ &= 0 + 0 + 0 + \sigma_\epsilon^2 \\ &= \sigma_\epsilon^2 \end{aligned}$$

because the first three terms are uncorrelated with ϵ_t.

ARMA(1,1): 2

Assuming weak stationarity, we have constant variances, σ_X^2, and:

$$\begin{aligned}
\sigma_X^2 &= Var(\phi_0 + \phi_1 X_{t-1} + \theta_1 \epsilon_{t-1} + \epsilon_t) \\
&= \phi_1^2 \sigma_X^2 + \theta_1^2 \sigma_\epsilon^2 + \sigma_\epsilon^2 + 2Cov(\phi_1 X_{t-1}, \theta_1 \epsilon_{t-1}) \\
&= \phi_1^2 \sigma_X^2 + (1 + \theta^2 + 2\phi_1\theta_1)\sigma_\epsilon^2
\end{aligned}$$

Solving for σ_X^2, we find

$$\sigma_X^2 = \frac{(1 + \theta^2 + 2\phi_1\theta_1)\sigma_\epsilon^2}{1 - \phi_1^2}.$$

ARMA(1,1): 3

- Having found the mean and variance of the ARMA(1,1), we move on to the autocovariance and autocorrelation functions. $\gamma(h)$ is found by multiplying the defining equation for the ARMA(1,1) process by X_{t-h} and taking expectations on both sides of the equation. The autocorrelation is then found by dividing by σ_X^2. It is easiest to set $\mu = \theta_0 = 0$ as the constant term will not change $\gamma(h)$. First, $\gamma(1) = E(X_t X_{t-1})$, and

$$
\begin{aligned}
E(X_t X_{t-1}) &= \phi_1 E(X_{t-1}^2) + \theta_1 E(X_{t-1}\epsilon_{t-1}) + E(X_{t-1}\epsilon_t) \\
&= \phi_1 \sigma_X^2 + \theta_1 \sigma_\epsilon^2 + 0 \\
&= \frac{(\phi_1(1 + \theta_1^2 + 2\phi_1\theta_1) + \theta_1(1 - \phi_1^2))\sigma_\epsilon^2}{1 - \phi_1^2} \\
&= \frac{(\phi_1 + \theta_1)(1 + \phi_1\theta_1)\sigma_\epsilon^2}{1 - \phi_1^2}
\end{aligned}
$$

ARMA(1,1): 4

We divide by σ_X^2 to find $\rho(1)$

- For $h = 1$, the result is

$$\rho(1) = \frac{(1+\phi_1\theta_1)(\phi_1+\theta_1)}{1+\theta_1^2+2\phi_1\theta_1}.$$

- For $h \geq 1$, the calculation is simple. Recall that for an MA(q) process, the autocorrelation function is 0 when $h > q$. In this case, the MA parameter is 1, so the autocorrelation function will have the same geometric decay that we observed in the AR(1) process.

ARMA(1,1): 5

Specifically, for $h \geq 2$, multiplying by x_{t-h} and taking expectations:

$$\gamma(h) = E(\phi_1 X_{t-1} X_{t-h} + \theta_1 X_{t-h}\epsilon_{t-1} + X_{t-h}\epsilon_t) = \phi_1 \gamma(h-1)$$

and

$$\rho(h) = \phi_1 \rho(h-1),\ h \geq 2.$$

- Having found $\rho(1)$, we conclude that for $h \geq 2$,

$$\rho(h) = \phi_1^{h-1} \rho(1).$$

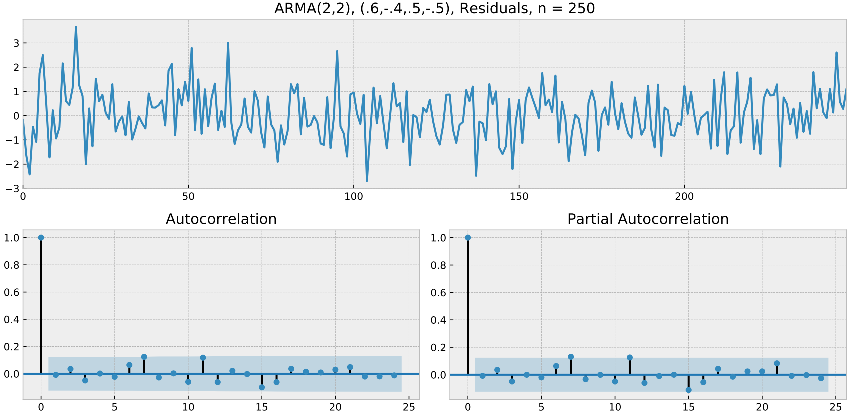
ARMA(p,q) Analysis: 1

```
# Simulate the ARMA series
arparams = np.array([.60, -.40]) # AR coefficients
ar = np.r_[1, -arparams] # add zero-lag and negate
maparams = np.array([.50,-.50]) # MA coefficients
ma = np.r_[1, maparams] # add zero-lag
arma_process = ArmaProcess(ar, ma)

arma_process.isstationary
True

arma_process.isinvertible
True
```

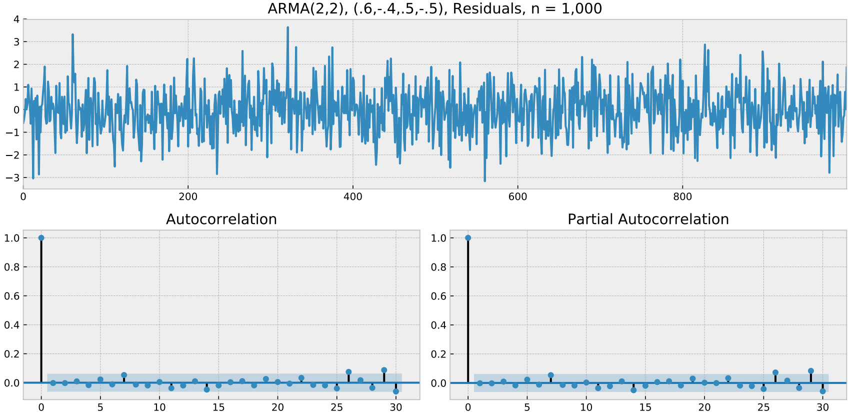
ARMA(p,q) Analysis: 2

```
# Fit the model
y = arma_process.generate_sample(250)
model = ARMA(y, (2, 2)).fit(trend='nc', disp=0)
array([ 0.657,-.305, .377, -.622]) #n = 250
array([.657,-.402,465,-.535]) # n = 1,000

print(model.params)
tsplot(model.resid)
plt.show()
```

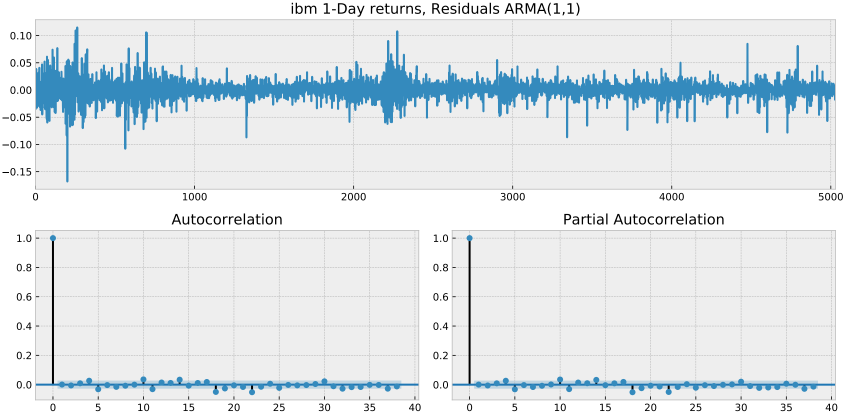
Residuals from ARMA(2,2), n = 250



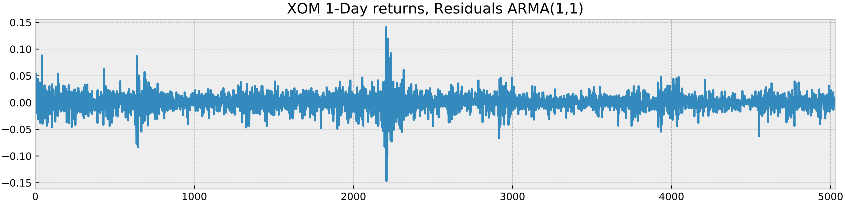
Residuals from ARMA(2,2), n = 1,000

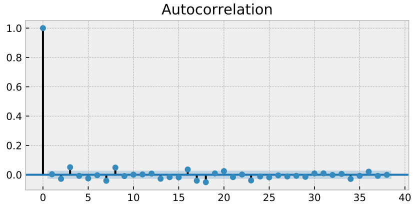


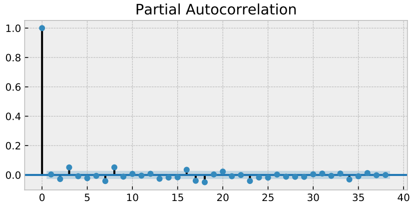
IBM One-Day Log Returns, ARMA(1,1) Residuals



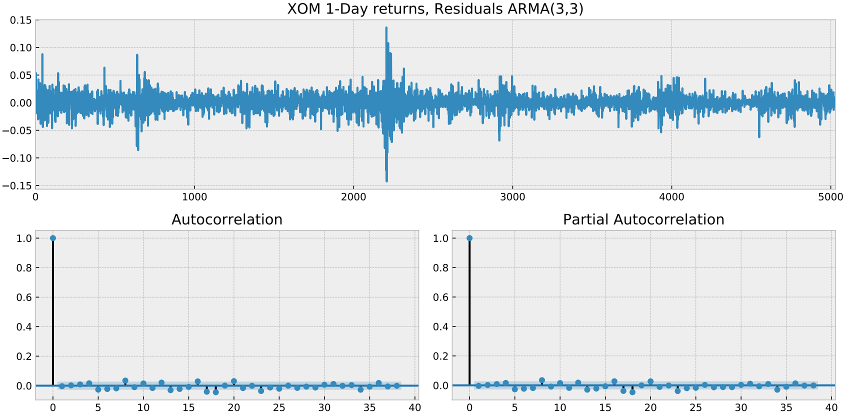
XOM One-Day Log Returns, ARMA(1,1) Residuals







XOM One-Day Log Returns, ARMA(3,3) Residuals



ARIMA and Differencing: 1

- There is yet a broader class of basic time series models, the ARIMA(p,d,q) models. Here the "i" in ARIMA stands for "integrated," and the middle parameter, d, stands for "difference."
- Recall problem #1 on Homework #2, that differencing a weakly stationary time series yields a new time series that is also weakly stationary.
- It is also possible that differencing a non-weakly stationary times series might transform it to become weakly stationary.

Arima and Differencing: 2

- A simple example of this is a random walk model. Suppose we begin with $\{\epsilon_t,\ t > 0\}$, a WN$(0, \sigma^2)$, and we create an AR(1) sequence with $\phi_1 = 1$

$$X_t = X_{t-1} + \epsilon_t.$$

- This is not a weakly stationary sequence as the characteristic polynomial has a root with modulus 1 (a unit root).
- Now consider the first difference sequence, $Y_t = X_t - X_{t-1}$. Note the Y_t is just the ϵ_t sequence, which ia weakly stationary by assumption.

Arima and Differencing: 3

- This simple example shows that taking differences of non-stationary time series is one approach to transforming a non-weakly stationary series into a weakly stationary one.
- Of course this works in only one direction, doing the opposite of differencing (integrating) can turn an otherwise weakly time series into a non-stationary one.
- Consequently, the d parameter indicates how many differences need to be applied in order to transform the time series so it is weakly stationary.

Differencing to obtain Stationarity: 1

- We introduce the "Backshift" or "lag" operator, B, and the differencing operator, ∇, defined by:

$$B^k(X_t) = B^k X_t = X_{t-k},\ k = 1, 2, \ldots,$$

$$\nabla X_t = X_t - X_{t-1} = X_t - BX_t = (1 - B)X_t.$$

- The last formula can be iterated to give

$$\nabla^k X_t = (1 - B)^k X_t.$$

- Note

$$\nabla^2 X_t = (1 - 2X_{t-1} + X_{t-2}).$$

Differencing to obtain Stationarity: 2

- Definition: A univariate time series $\{X_t\}$ is integrated of order k (denoted by $\mathcal{I}(k)$) if $\nabla^{k-1} X_t$ is non-stationary (hence $\nabla^{k-i} X_t$ is non-stationary for $i = 1, \cdots, k-1$); however $\nabla^k X_t$ is stationary.
- An $m-$dimensional time series is integrated of order k (denoted $\mathcal{I}(k)$), if at least one of its coordinates is $\mathcal{I}(k)$ and all the others are $I(j)$ for some $j \leq k$.

Testing for Stationarity
Dickey-Fuller Test

- Begin with the AR(1) process, $X_t = \phi_1 X_{t-1} + \epsilon_t$, with ϵ_t assumed to be a stationary process. The $\{X_t\}$ process is stationary if and only if $|\phi_1| < 1$.
- Test: $H_0 : |\phi_1| = 1$ versus $H_1 : |\phi_1| < 1$.
- There are two approaches:
    1. Regress X_t on X_{t-1} and set up a “t-statistic,”
    2. Subtract X_{t-1} from both sides, so $\nabla X_t = (\phi_1 - 1)X_{t-1} + \epsilon_t$, regress ∇X_t on X_{t-1}, and set up a “t-statistic”.

Dickey-Fuller Test: 2

- Both approaches give identical estimates of β and identical “t-statistics”. To test for the unit root, we need the distribution of the test statistics under $H_0 : \phi_1 = 1$. In this case, the time series behaves like a random walk, or for long series, when properly scaled like a Wiener process. This gives rise to the Dickey-Fuller test.

Dickey-Fuller Test: 3

$$\hat{\phi}_1 = \frac{\sum_{t=1}^T X_{t-1} X_t}{\sum_{t=1}^T X_{t-1}^2}$$

$$\hat{\phi}_1 - 1 = \frac{\sum_{t=1}^T X_{t-1}(X_t - X_{t-1})}{\sum_{t=1}^T X_{t-1}^2}$$

$$s_T^2 = \frac{1}{T-1} \sum_{t=1}^T (X_t - \hat{\phi}_1 X_{t-1})^2$$

$$t_T = \frac{\hat{\phi}_1 - 1}{s_T / \sqrt{\sum_{t=1}^T X_{t-1}^2}}$$

Under H_0, $\phi_1 = 1$ and the time series is a random walk. For large samples, both numerator and denominator can be approximated using central limit theorem ideas.

Dickey-Fuller Test: 4

$\phi_1 = 1, T \to \infty$

$$Y_t = \frac{X_t}{\sigma\sqrt{T}} \Rightarrow W(\frac{t}{T}) = W(s),\ s = \frac{t}{T},\ 0 \le s \le 1.$$

$$T(\hat{\phi}_1 - 1) = \frac{\sum_{t=1}^{T} Y_{t-1}(Y_t - Y_{t-1})}{\sum_{t=1}^{T} Y_{t-1}^2 \frac{1}{T}} \Rightarrow \frac{\int_o^1 W(s)dW(s)}{\int_0^1 W^2(s)ds}.$$

$$t_T = \frac{\hat{\phi}_1 - 1}{s_T/\sqrt{\sum_{t=1}^{T} X_{t-1}^2}}$$

$$t_T \Rightarrow \frac{\int_o^1 W(s)dW(s)}{\sqrt{\int_0^1 W^2(s)ds}} = \frac{\frac{1}{2}(W^2(1) - 1)}{\sqrt{\int_0^1 W^2(s)ds}}.$$

Dickey Fuller Test: 5

- The limiting probability distribution is different from the t-distribution. For example $P(t_T < 0) = P(W^2(1) < 1) = P(-1 < Z < 1) = .68$, so the distribution is not symmetric, unlike the t-distribution.
- This distribution is tabled, and used in statistical packages. Remember that it is derived under certain specific assumptions. The analysis is correct for $T \to \infty$ as long as the ϵ_t has finite second moments. However, in smaller samples and heavy tailed distributions, the distribution will not be correct.
- Can use Monte Carlo methods to simulate the distribution of t_T under any specific model.

Dickey Fuller Test: 6

```
import numpy as np
import statsmodel.tsa.api as smt

n = 1000
phi0 = 0
phi1 = 1
sigma = 1
z = sigma*np.random.randn(n)
x[0] = sigma*z[0]
for i in range(n-1):
```

$$x[i+1] = phi0 + phi1 * x[i] + z[i+1]$$

```
smt.statools.adfuller(x)
(-0.96, 0.77, 0, 999, ('1%': -3.44, '5%': -2.86, %'10%': -2.57))
smt.stattools.adfuller(z)
(-23.75, 0.0, 1, 998, ('1%': %-3.44, '5%': -2.86, %'10%': -2.57))
```